Tukue Gebremariam Gebregergis

Senior Backend and Platform Engineer

Java • Python • Cloud Infrastructure • DevOps • Applied AI

Stockholm, Sweden | +46 725838209 | tukue.geb@gmail.com
github.com/tukue

Professional profile

Backend and platform engineer with experience building Java microservices, automating cloud infrastructure and improving software delivery. Professional background spans telecom, media, public transport and digital document technology, with hands-on work in AWS, Kubernetes, CI/CD and infrastructure as code. Current independent work focuses on full-stack applications, agentic AI and developer workflows, alongside continued development in Azure and GCP.

Independent development and projects

November 2024 – Present | Independent projects and professional development

Building and maintaining personal applications while strengthening backend, platform engineering and AI engineering skills. This work complements previous employed roles and provides a practical setting for exploring architecture, automation and AI-assisted development.

Agentic AI application
Independent project

- Exploring AI agents and agent-based application workflows through hands-on development.
- Working with AI coding assistants, project context and instructions in personal development workflows.
- Deepening knowledge of Model Context Protocol (MCP), LLMOps and MLOps concepts.

Full-stack e-commerce application
Independent project | React, TypeScript, Node.js, PostgreSQL

- Developing a full-stack application spanning the user interface, backend services and relational data storage.
- Maintaining and iterating on the deployed application through ongoing development.

Cloud infrastructure and delivery automation
Independent projects | Terraform, CloudFormation, Bicep, CI/CD

- Building infrastructure-as-code solutions and extending practical skills in Kubernetes, GitOps and DevSecOps.
- Applying cloud and automation experience to reusable development and deployment workflows.

Professional experience

Enigio | Senior Cloud Engineer
May 2024 – October 2024

- Designed and implemented an event-driven application in JavaScript using AWS services, including Lambda.
- Automated ECS container restarts in response to security patching events and provisioned infrastructure with CloudFormation.
- Worked with AWS Security Hub and Inspector to support cloud security scanning.

Digital Route | DevOps Engineer / SRE
June 2023 – October 2023

- Developed infrastructure as code using Terraform and worked with AWS services and Kubernetes operations.
- Conducted AWS Well-Architected Reviews of production environments and automated security practices.
- Monitored production environments using Datadog.

Knowit / Trafikförvaltningen | Cloud Developer Consultant
January 2022 – May 2023

- Implemented cloud migration and infrastructure automation initiatives.
- Developed serverless AWS applications and IoT systems, and implemented CI/CD pipelines using Azure DevOps.
- Migrated databases using AWS Database Migration Service.

Telenor | Backend Developer
December 2017 – January 2022

- Developed REST APIs and CRM microservices using Java, Spring Boot and Hibernate.
- Integrated TMForum interoperability standards and implemented DevSecOps practices in CI/CD and Kubernetes environments.
- Worked with Grafana, Prometheus and Elasticsearch for application monitoring.

Discovery Network Sweden | System Development Engineer
March 2016 – December 2017

- Developed Java integrations for TV planning systems and migrated applications to AWS.
- Performed troubleshooting, maintenance and REST API integration testing.

Ericsson AB | LTE Software Developer
August 2014 – December 2015

- Developed Linux-based embedded software in C for 4G HetNet features.
- Conducted end-to-end testing and feature verification in an Agile development environment.

Technical skills

Backend and programming: Java, Spring Boot, Hibernate, Python, Flask, Node.js, JavaScript, TypeScript, C and C++.

Frontend: React, HTML, CSS and Bootstrap.

Cloud and infrastructure: AWS, Azure, GCP, Kubernetes, Docker, Terraform, CloudFormation, Bicep and Ansible.

Delivery and automation: Git, GitHub Actions, Azure DevOps, Jenkins, AWS CodeBuild, AWS CodePipeline, GitOps and Argo CD.

Data and observability: PostgreSQL, DynamoDB, MongoDB, Oracle, Datadog, Prometheus, Grafana and Elasticsearch.

AI engineering focus: Generative AI, AI-assisted development, agentic workflows, MCP, and continued hands-on learning in MLOps and LLMOps.

Education

MSc in ICT Innovation
KTH Royal Institute of Technology, Sweden | 2013–2014

- Communication System Design specialization; thesis research on energy-efficient base station operation.

MSc in Computer Networks
EIT Digital Master Program, France | 2012–2013

- University Pierre et Marie Curie and ParisTech; focus on internet technology and network architecture.

BSc in Electronics and Communication Engineering
Mekelle University, Ethiopia | 2005–2009

- Graduated with Great Distinction.

Further training

Fullstack and Python Development Bootcamp | October 2023 – May 2024
Built backend APIs and full-stack applications using Flask, Node.js, React and TypeScript.

AI for Technical Leaders | Quantic — completed
Fullstack Software Developer Java AI | Lexicon — ongoing

Certifications

AWS Certified Solutions Architect – Associate • AWS Certified Developer – Associate • AWS Certified AI Practitioner • Microsoft Certified Azure Fundamentals • Terraform Associate • ISTQB Certified Tester Foundation Level • Certified Scrum Product Owner

Languages

English: professional proficiency. Swedish: conversational; continuing language development.